

FORMULAE:

1. Measures of central tendency

* Sample / population Mean (Raw Data):

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i = \frac{\sum x}{n}$$

* Mean from a frequency table:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

* Median position (Individual Data):

• For odd n :

$$\text{position} = \frac{n+1}{2}$$

• For even n :

$$\text{Median} = \frac{\text{value at } (n/2) + \text{value at } (n/2 + 1)}{2}$$

* Symmetric Distribution property:

$$\text{Mean} = \text{Median} = \text{Mode}$$

2. Measures of Dispersion and outlier Detection

* Range:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

* position / sample variance (σ^2):

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

- Standard Deviation (σ):

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} = \sqrt{\text{variance}}$$

- Quartile positions (ordered data):

- First Quartile (Q_1 / 25th percentile)

$$\text{position} = \frac{n+1}{4}$$

- Second Quartile (Q_2 / 50th percentile (Median))

$$\text{position} = \frac{n+1}{2}$$

- Third Quartile (Q_3 / 75th percentile)

$$\text{position} = \frac{3(n+1)}{4}$$

- Fourth Quartile (Q_4 / 100th percentile (Max))

Value at position n (or $n+1$ by percentile formula).

- Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

- IQR Outlier Detection Bounds:

$$\text{Lower Bound} = Q_1 - 1.5(IQR)$$

$$\text{Upper Bound} = Q_3 + 1.5(IQR)$$

- Outlier condition: $x < \text{Lower Bound}$ or $x > \text{Upper Bound}$

3. Feature Scaling and Transformations:

- Standardization (z-score):

$$z = \frac{x - \mu}{\sigma}$$

- Outlier rule for normal distribution:
flag as outlier if $|z| > 3$

- Min max Normalization (scaling to $[0, 1]$):

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

4. Covariance and correlation

- Sample Covariance:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

- Pearson Correlation Coefficient (r):

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

- Range: $-1 \leq r \leq 1$

5. Probability Rules and Theorems

- Basic classical Probability:

$$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}, 0 \leq P(A) \leq 1$$

- Complement Rule:

$$P(A^c) = 1 - P(A) \text{ or } P(A) + P(A^c) = 1$$

- Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- For mutually exclusive events $P(A \cap B) = 0$:

$$P(A \cup B) = P(A) + P(B)$$

- Multiplication Rule

$$P(A \cap B) = P(A) \cdot P(B|A)$$

- For independent events ($P(B|A) = P(B)$):

$$P(A \cap B) = P(A) \cdot P(B)$$

- Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Baye's Theorem and Law of Total probability:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

- Expanded form:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

6. Probability Distributions:

- Continuous Probability Density Function (PDF) and Area:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- Total Area Requirement:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

- Normal (Gaussian) Distribution PDF:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Empirical 68-95-99.7 Rule:

- $\mu \pm 1\sigma \approx 68\%$ of data

- $\mu \pm 2\sigma \approx 95\%$ of data

- $\mu \pm 3\sigma \approx 99.7\%$ of data

- Binomial Distribution:

$$P(X=r) = {}^n C_r p^r q^{n-r}$$

where

n = number of trials

r = number of successes

p = probability of success

$q = 1-p$ = probability of failure

- Poisson Distribution:

$$P(X=n) = \frac{e^{-\lambda} \lambda^n}{n!}$$

Where

n = number of events (0, 1, 2, 3, ...)

λ = average rate of occurrence in the interval ($\lambda > 0$)

$$e \approx 2.718$$

7. Hypothesis Testing

- Standard Error of the Mean (SE / $\sigma_{\bar{n}}$):

$$\sigma_{\bar{n}} = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad s_{\bar{n}} = \frac{s}{\sqrt{n}}$$

- One-sample Z-Test (known σ):

$$Z = \frac{\bar{n} - \mu_0}{\sigma / \sqrt{n}}$$

- One sample t-test (Unknown σ , small sample):

$$t = \frac{\bar{n} - \mu_0}{\frac{s}{\sqrt{n}}}, \quad df = n - 1$$

Where sample standard-deviation $s = \sqrt{\frac{\sum (\bar{n} - \bar{n})^2}{n-1}}$

- Two-sample Independent t-test (Equal variances):

$$t = \frac{\bar{n}_1 - \bar{n}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}, \quad df = n_1 + n_2 - 2$$

- paired t-Test:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}, \quad df = n-1$$

- One sample proportion z-Test:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0 \left(\frac{1-p_0}{n} \right)}}$$

- Confidence Interval for the Mean:

- When σ is known:

$$CI = \bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

- When σ is unknown:

$$CI = \bar{x} \pm t_{\alpha/2, df} \left(\frac{s}{\sqrt{n}} \right)$$

- Chi square Test for Independence / Goodness of Fit:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \quad df = (r-1)(c-1) \text{ or } (k-1)$$